A Leakage-Free Machine Learning Pipeline for Cardiovascular Disease Prediction from Imbalanced Survey Data

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ABSTRACT

Cardiovascular disease (CVD) remains one of the foremost causes of death globally, and machine learning (ML) models trained on population health surveys are widely proposed for early risk screening. Because CVD cases are a minority in such data, studies commonly rebalance the classes with hybrid sampling (HS), here SMOTEENN followed by Random Oversampling (ROS). This study evaluates HS with three deep learning (DL) models (CNN, NN, MLP) and three classical ML models (Decision Tree, SVM, XGBoost) on 229,781 unique records of the 2015 BRFSS survey, with 6,000 real test records held out before any sampling. On these unseen records the HS models reach only 57.4–70.8% accuracy and an area under the ROC curve (AUC) of 0.57–0.81, and they miss most CVD cases (recall 18.5–60.2%): by deleting borderline records and duplicating the remaining minority records, HS leaves only the most clear-cut CVD cases in the training data. We therefore propose a pipeline that uses all 21 survey features, class weighting instead of synthetic resampling, regularised models and a decision threshold tuned on a validation set. It improves every model on unseen patients, to 74.9–76.0% accuracy, 78.8–82.8% recall and an AUC of 0.825–0.840 (Decision Tree accuracy 57.4% → 74.9%, XGBoost recall 30.5% → 82.8%), with five-fold cross-validation confirming these values (AUC 0.82–0.84) and bootstrap confidence intervals showing every gain to be significant. After a simple correction for class weighting, the predicted risks are well calibrated (Brier score 0.070–0.072 vs. 0.084 for the prevalence alone), and the strongest risk factors are a history of stroke, male sex, high cholesterol and high blood pressure. Simple models such as logistic regression perform on par with deep networks. On the same BRFSS data the improved pipeline exceeds the AUC of a published LightGBM model (0.840 vs. 0.811), and applied unchanged to the clinical Cleveland dataset it reaches 80–81% accuracy and an AUC of 0.89–0.90. The results show that, for CVD screening from imbalanced survey data, how class imbalance is handled matters more than the choice of model, and they provide realistic expectations and a reproducible, open-source baseline.

Keywords: Cardiovascular Disease, Disease Prediction, Machine Learning, Neural Networks, Hybrid Sampling, Class Imbalance, Data Leakage

1 INTRODUCTION

According to the World Health Organization (WHO), CVDs are a group of conditions that affect the heart and blood vessels, leading to various health issues and accounting for approximately 31% of all global deaths annually [1]. It has been estimated that by 2030, deaths related to CVD will rise significantly, with approximately 23.3 million people dying from this condition [2]. These types of diseases include, but are not limited to, coronary artery disease, myocardial infarctions, angina, stroke, arrhythmias, atherosclerosis, and hypertension. Myocardial infarctions and angina are primarily caused when the arteries become narrowed or blocked, resulting in reduced blood supply to the heart, thereby escalating the risk of CVDs [3, 4]. It has also been found that high blood pressure, along with arterial stiffening, is a major cause of CVD. Additionally, factors such as alcohol use, stress, physical inactivity, obesity and its complications, and diabetes also contribute to the development of CVD [5]. Given the silent yet deadly progression of heart disease, early and accurate diagnosis becomes increasingly challenging. Therefore, early detection and intervention in CVD are crucial, as timely and accurate treatment can help prevent life-threatening complications [6].

Although age, gender, blood pressure, and smoking status have traditionally been employed as fixed factors for predicting CVD, they do not adequately reflect the disease's dynamic nature and complexity [7, 8]. In contrast, techniques in data mining and machine learning (ML) have demonstrated encouraging results in assessing the severity of heart disease and forecasting outcomes, with algorithms such as K-nearest neighbors (KNN), Decision Tree (DT), and Naive Bayes (NB) showing effectiveness [9]. Consequently, advanced analytical approaches are crucial not only for early detection but also for the effective management of the numerous factors linked to CVD. However, deep learning (DL) approaches often encounter challenges concerning data interpretation, transparency, and reliability in medical contexts [10]. To overcome these limitations, hybrid models combining DL with other computational techniques have been developed, offering enhanced explainability, predictive accuracy, and clinical relevance. Such approaches have been applied across cardiovascular and related health problems, for example to study how peripheral muscle action limits exercise capacity in children with congenital heart defects [11], to monitor fetal health with hybrid DL algorithms [12], and, combined with blockchain, to secure e-healthcare data [13].

A second, less discussed challenge is class imbalance. In population health surveys only a small fraction of respondents report CVD (about 10% in the data used here), so classifiers trained naively tend to predict the majority class. Resampling methods such as the Synthetic Minority Oversampling Technique (SMOTE), Edited Nearest Neighbours (ENN) and Random Oversampling (ROS), alone or combined into hybrid sampling (HS), are widely used to rebalance the classes [14, 15, 16]. How and when resampling is applied, however, determines whether the reported performance reflects the model's ability to recognise new patients. If the full dataset is resampled before it is split into training and test sets, duplicated and synthetic records derived from the same patient can appear on both sides of the split, and cleaning steps can remove difficult cases from the test set. The resulting estimates can be far higher than the performance a clinic would observe.

This study examines this problem for CVD prediction from population survey data. We combine three DL models (CNN, NN, MLP) and three ML models (DT, SVM, XGBoost) with correlation-based feature selection and HS, and evaluate them with a leakage-free protocol in which the test data are held out before any sampling. We then identify which design choices limit performance on real patients, propose an improved pipeline, and compare it with published studies on the same and on clinical data. A comprehensive list of abbreviations and their corresponding descriptions can be found in Table 1. The key contributions of this study are as follows:

- We implement a typical HS pipeline for CVD prediction, hereafter called the baseline HS

pipeline, with data preprocessing, feature selection based on correlation, hybrid sampling (SMOTEENN followed by ROS) and six DL and ML models, and evaluate it on 6,000 real records held out before any sampling (57.4–70.8% accuracy, AUC 0.57–0.81).

- We analyse why hybrid sampling degrades performance on real patients: it removes borderline records and duplicates the remaining minority records, so that, for example, the share of CVD cases with a past stroke rises from 16.4% to 38.1% and the models miss most real CVD cases.
- We propose an improved, leakage-free pipeline (all features, class weighting, regularised models, validation-tuned decision threshold) that raises every model to about 76% accuracy, 80% recall and an AUC of 0.84, confirmed by five-fold cross-validation, and an ablation study that attributes the gain to each change.
- We quantify the uncertainty of all results with bootstrap confidence intervals, report adjusted odds ratios for the 21 survey variables, and show that class-weighted models need a simple prior correction to provide calibrated risk estimates.
- We compare the improved pipeline with published studies on equal terms: on the same BRFSS data it exceeds the AUC of a published model (0.840 vs. 0.811), and applied unchanged to the clinical Cleveland dataset it reaches an AUC of 0.89–0.90.
- The complete code is released as open source so that every number and figure in this paper can be reproduced.

The primary goal of this research is to provide a reliable assessment of how well ML models can identify individuals with CVD from survey data, and to give practical guidance on building and evaluating such models. The structure of this paper is as follows: Section 2 offers an overview of the related literature. Section 3 explains the methodology, including the evaluation protocol, the improved pipeline and the external evaluation on clinical data. Section 4 describes the models and evaluation metrics. Section 5 presents and discusses the results, and Section 6 concludes with a summary of the key findings and an analysis of their implications.

2 LITERATURE REVIEW

The literature relevant to this study falls into three strands: ML models for CVD prediction, methods for handling class imbalance, and the evaluation pitfalls that arise when the two are combined.

2.1 Machine Learning for CVD Prediction

Many studies have applied classical ML to clinical heart disease data. Li et al. [18] investigated ML techniques for detecting heart disease within e-healthcare frameworks, combining classifiers such as logistic regression and KNN with feature selection to extract the most relevant features, and reported promising accuracy and precision on a real-world dataset. Ahmad et al. [20] assessed logistic regression, random forests and SVM for the swift diagnosis of CVD and showed that hyperparameter tuning with GridSearchCV improves diagnostic accuracy. Ramesh et al. [21] developed predictive models with logistic regression, DT and SVM and evaluated them on real-world data with precision, recall and F1-score, emphasising that the algorithm should be chosen to suit the characteristics of the data. Population survey data have also been used: Deng et al. [42] trained an interpretable LightGBM model on the BRFSS 2015 survey used in this study and, with SHAP analysis, identified age, smoking status, diabetes, hypertension and high cholesterol as the most influential predictors.

Table 1: List of Abbreviations

Abbreviation	Description
WHO	World Health Organization
CVD	Cardiovascular Disease
DL	Deep Learning
CKD	Chronic Kidney Disease
HRFLM	Hybrid Random Forest with a Linear Model
ML	Machine Learning
MLP	Multi-Layer Perceptron
CNN	Convolutional Neural Network
NN	Neural Network
DT	Decision Tree
SVM	Support Vector Machine
KNN	K-Nearest Neighbor
NB	Naive Bayes
LR	Logistic Regression
HDPM	Heart Disease Prediction Model
SMOTE	Synthetic Minority Oversampling Technique
ENN	Edited Nearest Neighbours
SMOTEENN	SMOTE followed by ENN cleaning
ROS	Random Oversampling
HS	Hybrid Sampling (SMOTEENN followed by ROS)
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CDC	Centers for Disease Control and Prevention
BRFSS	Behavioral Risk Factor Surveillance System
ReLU	Rectified Linear Unit
PPV	Positive Predictive Value
HS-MLP, HS-CNN, HS-NN	HS combined with MLP, CNN, NN
HS-DT, HS-SVM, HS-XGBoost	HS combined with DT, SVM, XGBoost

Several studies combine multiple models or feature selection steps. Mohan et al. [28] integrated a hybrid random forest with a linear model (HRFLM) and reported an accuracy of 88.4%, outperforming the individual algorithms. Kavitha et al. [29] introduced a hybrid of Random Forest and DT that achieved an accuracy of 88.7%. Ali et al. [17] integrated the Relief and LASSO feature selection algorithms with an ensemble of classifiers to isolate the most clinically influential risk indicators, improving sensitivity and specificity. In the field of clinical decision support, Fitriyani et al. [19] introduced the Heart Disease Prediction Model (HDPM), which balances the training data with SMOTE-ENN before classification with XGBoost, and reported an accuracy of 95.90% on the Cleveland heart disease dataset.

Hybrid and deep models have also been proposed for related clinical problems. A DNN incorporating Glomerular Filtration Rate and kidney injury markers surpassed traditional ML approaches for the early detection of Chronic Kidney Disease (CKD) [24], and hybrid parallel classifiers [25], two-phase feature selection with adaptive SVM [26] and further hybrid approaches [27] have been developed for chronic disease classification. Beyond classification, Qureshi et al. [22] optimised the transfer of kinetic energy from aortic blood flow in mechanical circulatory support devices through computer modelling and exhaustive search, showing a potential route to higher cardiac output in heart failure patients. In their systematic review of ML for CVD prediction from medical big data, Azmi et al. [23] concluded that data quality, model interpretability and scalability must be addressed before such technologies can be widely implemented.

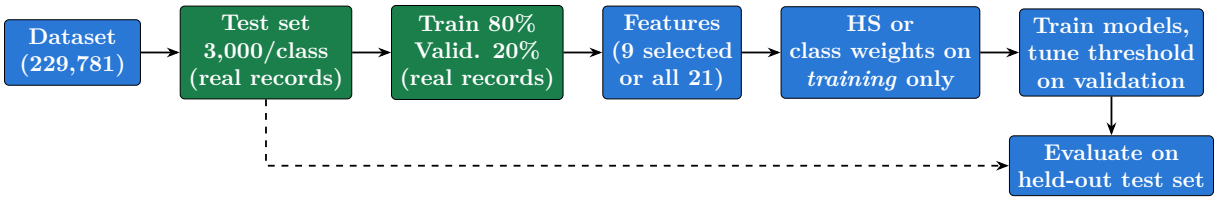


Figure 1: Overview of the methodology. The test set and the validation set are drawn from the real records before any sampling and are never resampled or used for training. The baseline HS pipeline uses the nine selected features and hybrid sampling of the training data; the improved pipeline uses all 21 features and class weights.

2.2 Class Imbalance and Resampling

Because CVD cases are usually a minority, class imbalance is a recurring challenge. Singh and Purohit [15] reviewed under-sampling, oversampling and hybrid techniques and stressed that the method should match the characteristics of the dataset. Batista et al. [16] compared these strategies, including SMOTE combined with ENN, and showed that balancing the training data can improve classifier performance; their work is a standard reference for hybrid sampling. In the cardiovascular domain, Ishaq et al. [14] used SMOTE to generate synthetic instances of the minority class for heart failure survival prediction and reported accuracies of 85% to 94% with KNN, NB, DT, XGBoost and Random Forests.

2.3 Data Leakage in Model Evaluation

Several authors have warned that preprocessing steps applied before data partitioning can leak information from the test set into model training and inflate reported performance [39, 40]. Kapoor and Narayanan [40] found such leakage in hundreds of published ML-based studies across many fields. Santos et al. [41] showed specifically that oversampling before cross-validation produces over-optimistic estimates on imbalanced data, and recommended that resampling be applied only to the training folds.

2.4 Research Gap

Hybrid sampling is widely recommended for imbalanced CVD data, and the studies above report accuracies of 85–96%. These figures, however, are obtained on different datasets, often small ones, and many studies do not state whether resampling or feature selection was applied before or after the data were split. It therefore remains unclear how much of the reported performance reflects genuine predictive power and how much is an artefact of the evaluation protocol. This study addresses this gap by evaluating a typical HS pipeline on real, held-out patients from a large survey dataset, analysing why it underperforms, and deriving an improved pipeline that is compared with published results on equal terms.

3 PROPOSED METHODOLOGY

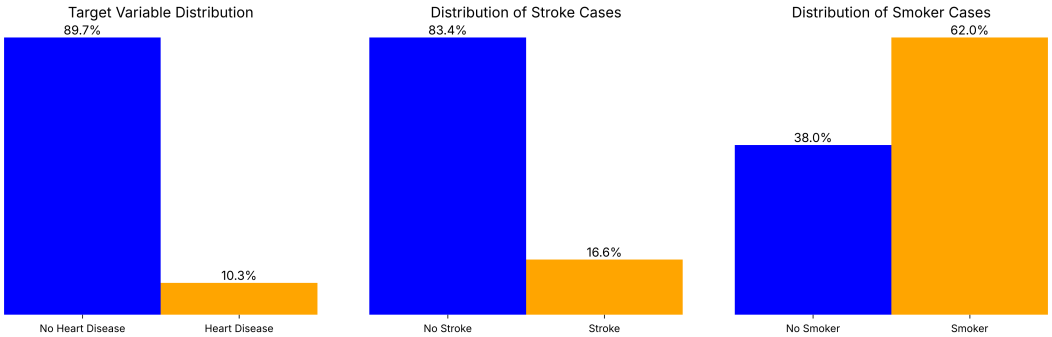
This section describes the data, the preprocessing and feature selection, hybrid sampling, the evaluation protocol, the improved pipeline and the external evaluation on clinical data. The analysis is implemented in Python using pandas, scikit-learn, imbalanced-learn, XGBoost and TensorFlow/Keras. The overall pipeline is shown in Fig. 1; its key design choice is the point at which sampling is applied relative to the train/test split: sampling is applied only to the training data, after the test data have been set aside.

Table 2: Description of Dataset Variables

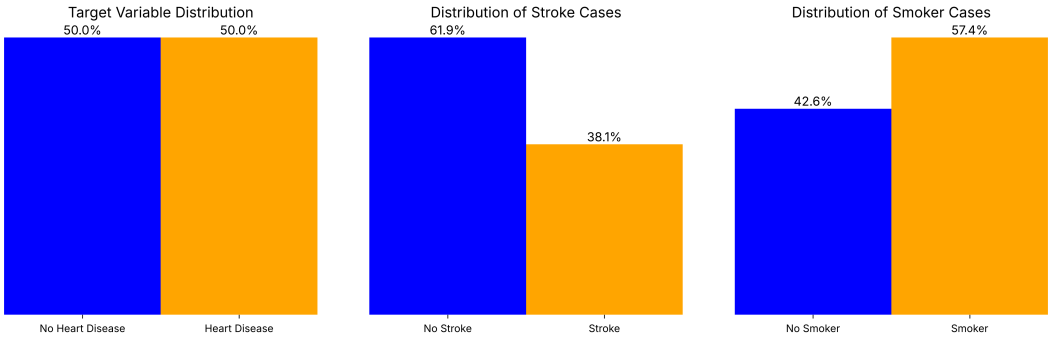
Variable	Description
HeartDiseaseorAttack	0 = No history of coronary heart disease (CHD) or myocardial infarction (MI); 1 = Reported CHD or MI
HighBP	0 = No high blood pressure; 1 = High blood pressure
HighChol	0 = No high cholesterol; 1 = High cholesterol
CholCheck	0 = No cholesterol check in past 5 years; 1 = Checked cholesterol in past 5 years
BMI	Body Mass Index [12–98]
Smoker	0 = Smoked fewer than 100 cigarettes in lifetime; 1 = Smoked at least 100 cigarettes in lifetime
Stroke	0 = Has never experienced a stroke; 1 = Has experienced a stroke
Diabetes	0 = Absence of diabetes; 1 = Prediabetes / borderline; 2 = Presence of diabetes
PhysActivity	0 = No physical activity or exercise in prior month; 1 = Engaged in physical activity or exercise in prior month
Fruits	0 = Consumes fruit less than once per day; 1 = Consumes fruit one or more times per day
Veggies	0 = Consumes vegetables less than once per day; 1 = Consumes vegetables one or more times per day
HvyAlcoholConsump	Adult men: $\geq$14 drinks/week; Adult women: $\geq$7 drinks/week (0 = No, 1 = Yes)
AnyHealthcare	Health care coverage / health insurance (0 = No, 1 = Yes)
NoDocbcCost	Could not see doctor due to cost in past 12 months (0 = No, 1 = Yes)
GenHlth	General health rating: [1, 2, 3] = Excellent, Very Good, Good; [4, 5] = Fair, Poor
MentHlth	Days of poor mental health in past 30 days [0–30]
PhysHlth	Days of physical illness or injury in past 30 days [0–30]
DiffWalk	Serious difficulty walking or climbing stairs (0 = No, 1 = Yes)
Sex	0 = Female, 1 = Male
Age	13-level category: 1 = 18–24, 2 = 25–29, 3 = 30–34, 4 = 35–39, 5 = 40–44, 6 = 45–49, 7 = 50–54, 8 = 55–59, 9 = 60–64, 10 = 65–69, 11 = 70–74, 12 = 75–79, 13 = 80 or older
Education	[1, 2, 3] = Did not graduate high school; 4 = High school graduate; 5 = Attended college or technical school; 6 = College graduate
Income	1, 2 = Lower than \$15,000; 3, 4 = \$15,000 to less than \$25,000; 5 = \$25,000 to less than \$35,000; 6 = \$35,000 to less than \$50,000; 7, 8 = \$50,000 or more

3.1 Data Collection

The dataset employed in this study was sourced from Kaggle [30], a publicly accessible data repository. It is a refined version of the data originally compiled by the Behavioral Risk Factor Surveillance System (BRFSS) [31]. The dataset consists of 253,680 records, each representing responses from the 2015 BRFSS survey conducted by the Centers for Disease Control and Prevention (CDC). It includes 22 columns, one of which is the target variable, "HeartDiseaseorAttack". This binary target variable indicates whether a respondent has ever been told they had coronary heart disease or a heart attack, where "0" signifies no history of coronary heart disease (CHD) or myocardial infarction (MI) and "1" indicates a reported history of CHD or MI. The raw data comprise 229,787 records without heart disease and 23,893 records with heart disease. A comprehensive description of the dataset, including the variables and their corresponding measurements, is provided in Table 2 [32].



(a) Before sampling



(b) Training set after hybrid sampling

Figure 2: Distribution of the target variable, and of stroke and smoker cases among individuals with heart disease, before and after hybrid sampling.

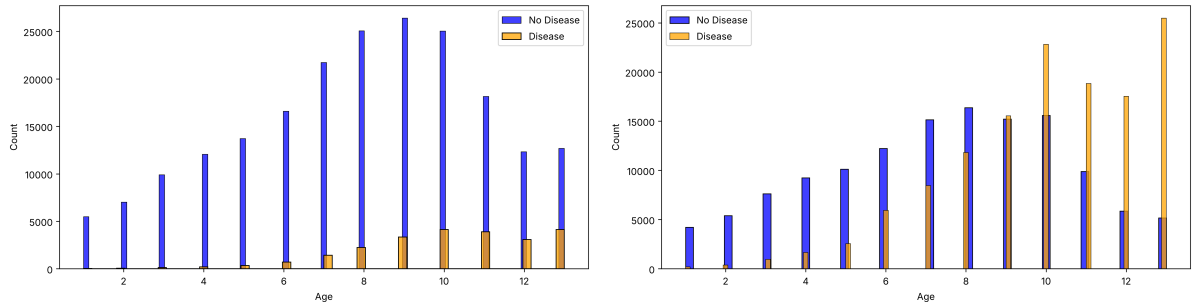
3.2 Data Preprocessing

The dataset is loaded with the pandas library and explored before modelling. Duplicate entries are removed to maintain data integrity, leaving 229,781 unique records with 22 columns. The presence of missing data is assessed, and the analysis confirms that no missing values exist. The target variable is strongly imbalanced: the dataset comprises 23,717 cases of heart disease (10.3%) and 206,064 instances of individuals without heart disease.

Fig. 2(a) depicts the distribution of the target variable and, among the heart disease cases, the proportions of individuals who have had a stroke (16.6%) or who smoke (62.0%). The age distribution of individuals with and without heart disease is compared in Fig. 3(a); the prevalence of heart disease increases steeply with age. Panels (b) of both figures show the data after hybrid sampling and are discussed in Section 3.4.

3.3 Feature Selection

Feature selection helps reduce the dataset's complexity while focusing on the variables most closely linked to the target outcome. A Pearson correlation matrix is computed and the features are ranked by their correlation with "HeartDiseaseorAttack". The strongest associations are found for GenHlth ($r = 0.246$), Age (0.224), DiffWalk (0.203), HighBP (0.201), Stroke (0.199), HighChol (0.176), Diabetes (0.171), PhysHlth (0.171) and Smoker (0.105). These nine features are used by the baseline HS models. Because Pearson correlation only captures linear, univariate association, the improved pipeline (Section 3.6) instead uses all 21 features and lets the models weigh them; the effect of this choice is quantified in the ablation study.



(a) Before sampling **(b)** Training set after hybrid sampling

Figure 3: Age distribution of individuals with and without heart disease before and after hybrid sampling.

3.4 Hybrid Sampling

Hybrid sampling combines SMOTEENN with ROS. SMOTE first synthesises new minority-class instances by interpolating between a minority record and its nearest minority neighbours until the classes are balanced. ENN then removes every record whose class disagrees with the majority of its three nearest neighbours, from both classes. Finally, ROS duplicates minority records until both classes have equal size. The HS methodology is implemented with the imbalanced-learn library (random state 42).

Applied to the 179,024 real training records (Section 3.5), SMOTE creates 145,876 synthetic positive records. ENN then removes 88.5% of the positive records together with 18.7% of the real negative records, and ROS duplicates each remaining positive record about seven times, giving 264,238 balanced training records. Hybrid sampling thus balances the classes (Fig. 2(b)), but it also changes who the records describe. Because ENN deletes records that resemble the opposite class, the remaining heart disease cases are the most clear-cut ones: among them, the share of individuals who have had a stroke rises from 16.4% to 38.1%. Conversely, the remaining non-CVD records are younger on average (mean age category 7.86 before and 7.38 after sampling; Fig. 3(b)), consistent with the removal of older healthy respondents who resemble patients. Section 5 shows how these changes affect performance on real patients.

3.5 Evaluation Protocol

Before any sampling, 3,000 *real* records per class are drawn from the deduplicated dataset as the test set. The remaining records are split, stratified by class, into training (179,024 records, 80%) and validation (44,757 records, 20%) sets. Sampling, when used, is applied to the training set only; the validation and test sets contain only real records with their natural characteristics, so that no synthetic, duplicated or cleaned record can reach the test data. The test set is balanced, so its accuracy equals the balanced accuracy. For cross-validation, the 223,781 non-test records are split into five stratified folds. Each held-out fold keeps the prevalence of the non-test records, 9.3% (slightly below the 10.3% of the full dataset because 3,000 CVD records were set aside for the balanced test set), and any sampling, early stopping and threshold tuning use only the training folds.

3.6 Improved Pipeline

Motivated by the ablation study in Section 5, we propose an improved pipeline that is evaluated with the same protocol. It makes four changes to the baseline HS pipeline:

1. **All 21 features** are used instead of the nine correlation-selected ones.

2. **Class weighting replaces resampling.** The models are trained on the real, imbalanced training data; errors on the minority class are weighted by the inverse class frequency (`class_weight="balanced"` for DT, SVM, LR and the DL models; `scale_pos_weight` = 9.8, the ratio of negative to positive training records, for XGBoost).

3. **Regularised models.** The DT is limited to a depth of 8 with at least 50 records per leaf; XGBoost uses shallow trees (depth 5), learning rate 0.03, row and column subsampling of 0.8 and early stopping on the validation AUC (patience 100 rounds, at most 2,000 trees); the DL models use early stopping on the validation loss (patience 3, at most 30 epochs, batch size 256) and restore the best weights. Logistic regression (LR) is added as an interpretable baseline.

4. **Decision threshold tuned on the validation set.** Instead of the default threshold of 0.5, the threshold that maximises Youden's index J = sensitivity + specificity − 1 on the validation set is used.

When the models are used to estimate the risk of an individual rather than to classify, their predicted probabilities are additionally corrected for the class weights (Section 4.4).

3.7 External Evaluation on Clinical Data

Most published CVD studies use the UCI Cleveland heart disease data [43], 303 hospital patients described by 13 clinical measurements, including chest-pain type, resting ECG, maximum exercise heart rate, exercise-induced ST depression and thallium scintigraphy. To compare with these studies on equal terms, the improved pipeline is applied, unchanged, to this dataset (PMLB version [44]). The 6 patients with missing values are removed, leaving 297, of whom 137 have heart disease, and the categorical variables are one-hot encoded. Because a single split of such a small dataset yields test sets of about 60 patients, the models are evaluated with five repetitions of stratified five-fold cross-validation; in every fold, early stopping and threshold tuning use only the training portion.

4 DL AND ML ARCHITECTURE ANALYSIS

4.1 ML Models

Decision Tree. A DT classifier (C4.5-style, entropy splitting criterion) learns a hierarchy of feature tests that map input features to the CVD label. In the HS pipeline the tree is grown to full depth.

SVM. An SVM classifier with a linear kernel separates the two classes with a maximum-margin hyperplane. Features are min–max scaled, and the model is fitted with the LIBLINEAR solver, which solves the same linear SVM problem as a kernel SVM with a linear kernel but scales to several hundred thousand records. Its decision values are used to compute the ROC curve.

XGBoost. XGBoost builds an ensemble of gradient-boosted trees, each correcting the errors of the previous ones. The HS pipeline uses the library defaults.

Logistic regression. LR models the log-odds of CVD as a linear function of the standardised features. It is used only in the improved pipeline, as an interpretable baseline.

The settings of all models in the improved pipeline (class weights, depth limit, early stopping and threshold tuning) are given in Section 3.6.

4.2 DL Models

All DL models share a preprocessing stage: features are min–max scaled and passed through an autoencoder whose encoder (one dense layer of 64 ReLU units, trained for 10 epochs to reconstruct its input with a sigmoid output layer and mean-squared-error loss) produces a 64-dimensional representation that is fed to the classifier. The classifiers are:

- **CNN**: Conv1D (8 filters, kernel size 2, ReLU) → max pooling (2) → flatten → dense (16, ReLU) → dense (1, sigmoid); 20 epochs.
- **NN**: dense (64, ReLU) → dense (32, ReLU) → dense (1, sigmoid); 10 epochs.
- **MLP**: dense (128) → dense (64) → dense (32), all ReLU → dense (1, sigmoid); 20 epochs.

All DL models are trained with the Adam optimiser and binary cross-entropy loss, with a batch size of 32 in the HS pipeline.

4.3 Evaluation Metrics

For performance evaluation, a confusion matrix is employed to analyze the results of each model. The matrix produces four key metrics: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). These metrics facilitate the computation of additional performance indicators.

1. Accuracy: the proportion of correct predictions [33]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

2. Precision (positive predictive value): the proportion of predicted CVD cases that are correct [34]:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

3. Recall (sensitivity): the proportion of actual CVD cases that are detected [34]:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

4. F1-score: the harmonic mean of precision and recall [33]:

$$\text{F1-score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

5. Area Under the Curve (AUC): the area under the receiver operating characteristic (ROC) curve, which traces the true positive rate (TPR) against the false positive rate (FPR) across classification thresholds [33]. The AUC ranges from 0.5 (random guessing) to 1.0 (perfect classification) and is computed from the continuous model scores:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}) \tag{5}$$

Unlike accuracy, precision and F1, the AUC does not depend on the decision threshold or on the class balance of the test set, which makes it the most robust metric for comparing models across studies.

4.4 Statistical Analysis

Confidence intervals. Uncertainty in the test-set results is quantified with a stratified bootstrap: the 3,000 positive and 3,000 negative test records are resampled with replacement 2,000 times, and 95% confidence intervals (CIs) are taken as the 2.5th and 97.5th percentiles. Differences between models are evaluated on the same resamples (paired bootstrap); a difference is considered significant when its 95% CI excludes zero.

Risk factors. To describe how each survey variable relates to CVD, an unweighted multivariable logistic regression with all 21 features is fitted to the real training records. We report odds ratios (ORs) per unit of each variable with 95% Wald CIs.

Calibration. Calibration is assessed on the predictions made for the held-out folds of the five-fold cross-validation, in which every non-test record is predicted once, at the natural prevalence, by a model that did not see it. We report the mean predicted risk, the Brier score (the mean squared difference between predicted probability and outcome) and calibration curves over ten quantile bins. Class weighting trains a model as if both classes were equally frequent, so its probabilities correspond to a prevalence of 50%. They are therefore corrected for the training prevalence π with

$$\frac{\hat{p}}{1-\hat{p}} = \frac{p}{1-p} \cdot \frac{\pi}{1-\pi}, \tag{6}$$

where p is the model output and $\hat{p}$ the corrected probability. The correction is monotonic, so it changes neither the AUC nor, with a correspondingly transformed threshold, any classification. The SVM is excluded because its outputs are not probabilities.

5 RESULTS AND DISCUSSION

The results are presented in eight parts. Section 5.1 evaluates the baseline HS pipeline on real, held-out records, and Section 5.2 identifies the design choices that limit its performance. Section 5.3 presents the improved pipeline with bootstrap confidence intervals, and Section 5.4 confirms its results with five-fold cross-validation. Section 5.5 describes the risk factors and Section 5.6 the calibration of the predicted risks. Section 5.7 compares the findings with existing studies, including the external evaluation on clinical data, and Section 5.8 discusses the limitations. All experiments use a fixed random state of 42 and can be reproduced with the released code.

5.1 Baseline HS Pipeline on Real, Unseen Records

Table 3 (left) shows the HS models trained on the resampled training set and evaluated on the 6,000 real records that were held out before sampling. They reach only 57.4–70.8% accuracy and an AUC of 0.57–0.81. The fully grown DT is close to chance level (AUC 0.574): it memorises the training records, many of them duplicates, and these do not recur among new patients. All HS models are also very conservative on real data: their recall ranges from 18.5% (DT) to 60.2% (SVM), so most CVD cases are missed. This follows from the changes that HS makes to the training data (Section 3.4): ENN keeps only the most clear-cut heart disease cases (38.1% with a past stroke, compared with 16.4% among the real training records), so the models learn that only such profiles are positive and classify most real patients with less typical answers as negative.

Table 3: Test performance on 6,000 real records (3,000 per class) held out before any sampling. Left: baseline HS pipeline (nine features, hybrid sampling of the training data, threshold 0.5). Right: improved pipeline (all 21 features, class weights, regularised models, threshold tuned on the validation set). All values in %.

	Baseline HS pipeline					Improved pipeline				
Model	Acc.	Prec.	Rec.	F1	AUC	Acc.	Prec.	Rec.	F1	AUC
DT	57.40	83.23	18.53	30.32	57.40	74.90	73.06	78.90	75.87	82.50
SVM	70.78	76.39	60.17	67.31	81.06	75.98	73.89	80.37	76.99	83.67
XGBoost	61.58	80.67	30.47	44.23	74.31	76.00	72.90	82.77	77.52	83.99
CNN	66.85	76.45	48.70	59.50	77.58	75.55	73.43	80.07	76.61	83.41
NN	64.48	78.31	40.07	53.01	74.67	75.88	74.46	78.80	76.57	83.85
MLP	63.05	80.90	34.17	48.04	72.53	75.47	72.37	82.40	77.06	83.79
LR	–	–	–	–	–	76.00	73.93	80.33	77.00	83.70

Table 4: Ablation study on the real test set. "Tuned" denotes the threshold that maximises Youden's J on the validation set.

Model	Configuration	AUC (%)	Acc. @ 0.5 (%)	Acc. @ tuned (%)
XGBoost	9 features, HS (baseline)	74.31	61.58	68.28
XGBoost	9 features, ROS only	82.06	74.35	74.25
XGBoost	9 features, no resampling	82.25	53.27	74.55
XGBoost	21 features, no resampling	83.54	54.75	75.25
XGBoost	21 features, class weights, regularised (improved)	83.99	76.18	76.00
DT	9 features, HS, full depth (baseline)	57.40	57.40	57.40
DT	21 features, class weights, depth 8 (improved)	82.50	74.93	74.90

5.2 Ablation Study

To identify which design choices limit performance on real data, we varied them one at a time for XGBoost and the DT (Table 4). The AUC measures how well a model ranks patients by risk, independently of the threshold; accuracy is reported both at the default threshold of 0.5 and at the threshold tuned on the validation set.

Four findings emerge. First, hybrid sampling is the main cause of the loss: replacing it with ROS alone raises the AUC of XGBoost from 0.743 to 0.821, and dropping resampling altogether gives the same AUC (0.823; difference from HS +7.9 percentage points, 95% CI 6.9–8.9), showing that synthetic records and ENN cleaning distort the data rather than add information. Second, without resampling the default threshold of 0.5 is unsuitable for imbalanced data (53.3% accuracy, because almost every record is predicted negative), whereas a threshold tuned on the validation set restores 74.6% accuracy; resampling and threshold tuning address the same problem, but tuning does so without altering the data. Class weights address it as well: with them, the default threshold is already close to the tuned one (76.18% vs. 76.00% accuracy for XGBoost), so in the improved pipeline threshold tuning mainly guards against a miscalibrated model. Third, using all 21 features instead of the nine correlation-selected ones increases the AUC from 0.823 to 0.835 (+1.3 percentage points, 95% CI 0.8–1.8): features such as sex and income carry predictive information that a univariate correlation ranking misses (Section 5.5). Fourth, regularisation (shallow, early-stopped boosting and a depth-limited tree) prevents the models from memorising individual records; together with the other changes, it turns the DT from a near-random classifier into a competitive model (AUC 0.825).

5.3 Improved Pipeline

Table 3 (right) shows that the improved pipeline raises every model on real, unseen records, and Fig. 4 summarises its results. Accuracy increases to 74.9–76.0%, recall to 78.8–82.8% and the

Table 5: AUC and recall (%) on the real test set with 95% bootstrap confidence intervals (2,000 stratified resamples), and the paired difference in AUC between the improved and the baseline HS model.

Model	Baseline HS pipeline AUC	Baseline HS pipeline Recall	Improved pipeline AUC	Improved pipeline Recall	ΔAUC
DT	57.4 [56.6, 58.2]	18.5 [17.2, 19.9]	82.5 [81.5, 83.5]	78.9 [77.4, 80.4]	+25.1 [24.1, 26.2]
SVM	81.1 [80.0, 82.2]	60.2 [58.4, 62.0]	83.7 [82.7, 84.7]	80.4 [78.9, 81.9]	+2.6 [2.1, 3.2]
XGBoost	74.3 [73.1, 75.5]	30.5 [28.8, 32.1]	84.0 [83.0, 85.0]	82.8 [81.4, 84.1]	+9.7 [8.6, 10.7]
LR	–	–	83.7 [82.7, 84.7]	80.3 [78.9, 81.8]	–
CNN	77.6 [76.5, 78.7]	48.7 [46.9, 50.5]	83.4 [82.4, 84.4]	80.1 [78.7, 81.6]	+5.8 [5.0, 6.6]
NN	74.7 [73.4, 75.9]	40.1 [38.3, 41.8]	83.9 [82.8, 84.8]	78.8 [77.3, 80.3]	+9.2 [8.2, 10.2]
MLP	72.5 [71.2, 73.8]	34.2 [32.5, 35.8]	83.8 [82.8, 84.8]	82.4 [81.0, 83.9]	+11.3 [10.2, 12.4]

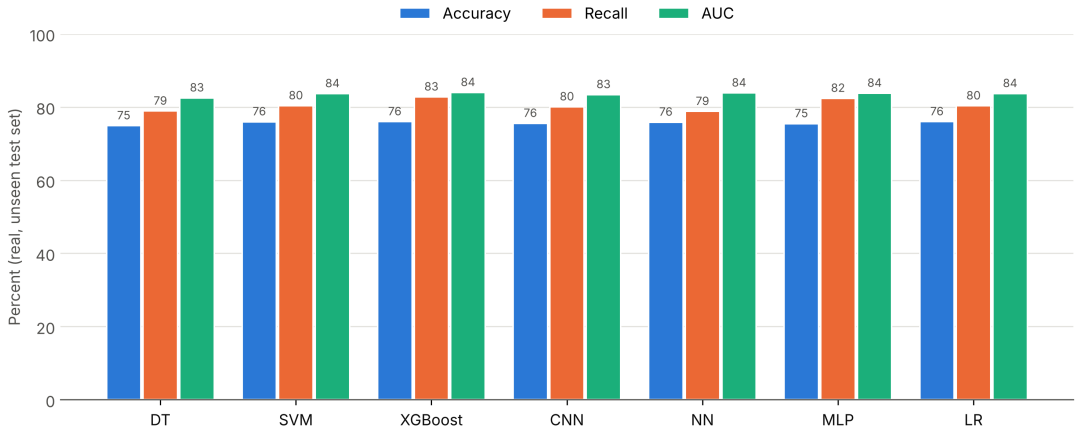


Figure 4: Accuracy, recall and AUC of the improved models on the real, unseen test set.

AUC to 0.825–0.840. The largest gains concern recall: the improved XGBoost detects 82.8% of CVD cases, compared with 30.5% for the baseline HS XGBoost, and the improved DT detects 78.9% compared with 18.5%. Table 5 gives 95% bootstrap CIs: every gain in AUC and recall is statistically significant, with CIs of the differences far from zero (for example, +9.7 AUC points, 95% CI 8.6–10.7, for XGBoost). The ROC curves of all improved models nearly coincide (Fig. 5). The improved XGBoost does not differ significantly from the NN and the MLP; its advantage over LR and the SVM (0.3 AUC points) is statistically significant but of no practical relevance, so logistic regression performs on par with XGBoost and the three DL models.

The confusion matrices (Fig. 6) show a balanced error profile: the improved XGBoost correctly identifies 2,483 of 3,000 CVD cases and 2,077 of 3,000 non-CVD cases. The learning curves of the DL models (Fig. 7) show the validation loss reaching its minimum after 6 (CNN), 11 (MLP) and 17 (NN) epochs and then levelling off; training stops three epochs after the last improvement and the weights of the best epoch are restored, which prevents over-fitting. The training loss is computed with class weights and is therefore not directly comparable to the unweighted validation loss.

5.4 Five-Fold Cross-Validation

To check that these results do not depend on one particular split, the improved models were evaluated with stratified five-fold cross-validation on the 223,781 real records that are not in the test set. In every fold, 20% of the training portion was set aside for early stopping and threshold tuning, and the held-out fold kept the natural prevalence of 9.3%. Table 6 reports the mean and standard deviation over the folds. The AUC (0.824–0.839) and recall (78.9–82.5%) agree with

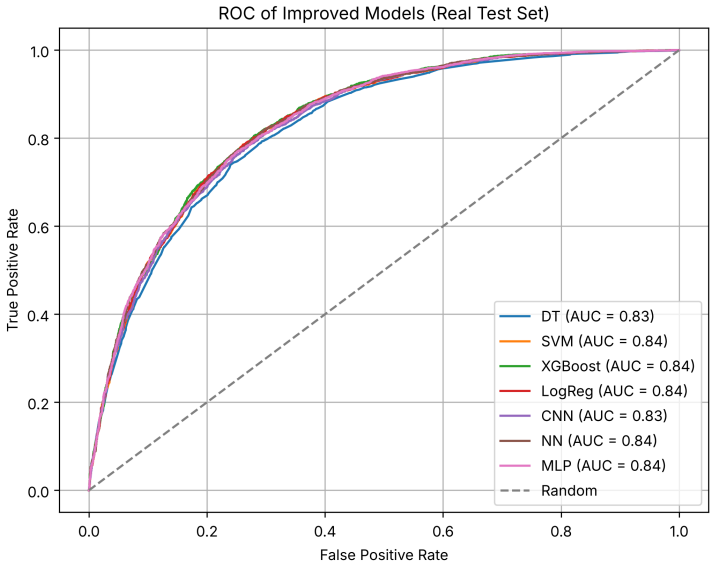
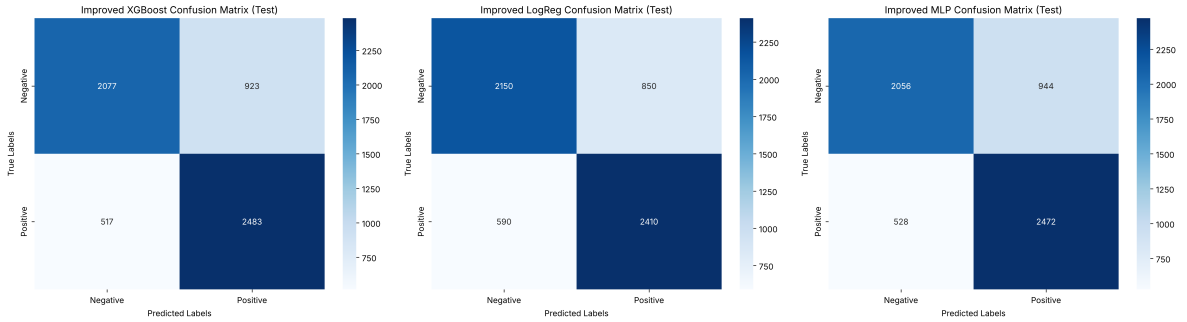


Figure 5: ROC curves of the improved models on the real, unseen test set.



(a) XGBoost **(b)** Logistic regression **(c)** MLP

Figure 6: Confusion matrices of improved models on the real test set (3,000 records per class).

the test-set results and vary little between folds (standard deviation of the AUC at most 0.64 percentage points), confirming that the estimates are stable. Accuracy (70.8–72.2%) is lower than on the balanced test set because the tuned threshold trades specificity for sensitivity and negatives make up about 91% of each fold.

Precision and F1 are much lower in cross-validation than on the balanced test set because the folds keep the real prevalence. At a prevalence of 10%, even a model with about 80% sensitivity and 70% specificity flags roughly three to four healthy respondents for every true CVD case (precision about 22%). This is an inherent property of screening for a condition of low prevalence, not a defect of a particular model, and it should guide how such models are used: as inexpensive pre-screening tools that refer at-risk individuals for clinical assessment, not as diagnostic tests.

5.5 Risk Factors

Table 7 lists the adjusted odds ratios of all 21 survey variables. The strongest associations with CVD are a history of stroke (OR 2.61, 95% CI 2.47–2.76), male sex (2.13), high cholesterol (1.84), a cholesterol check in the past five years (1.81), high blood pressure (1.67), poorer general health (1.60 per level), smoking (1.41), difficulty walking (1.34) and age (1.28 per five-year category). These factors agree with those identified on the same data by Deng et al. [42] (age, smoking, diabetes, hypertension and high cholesterol). Two of them, sex and the cholesterol check, are not

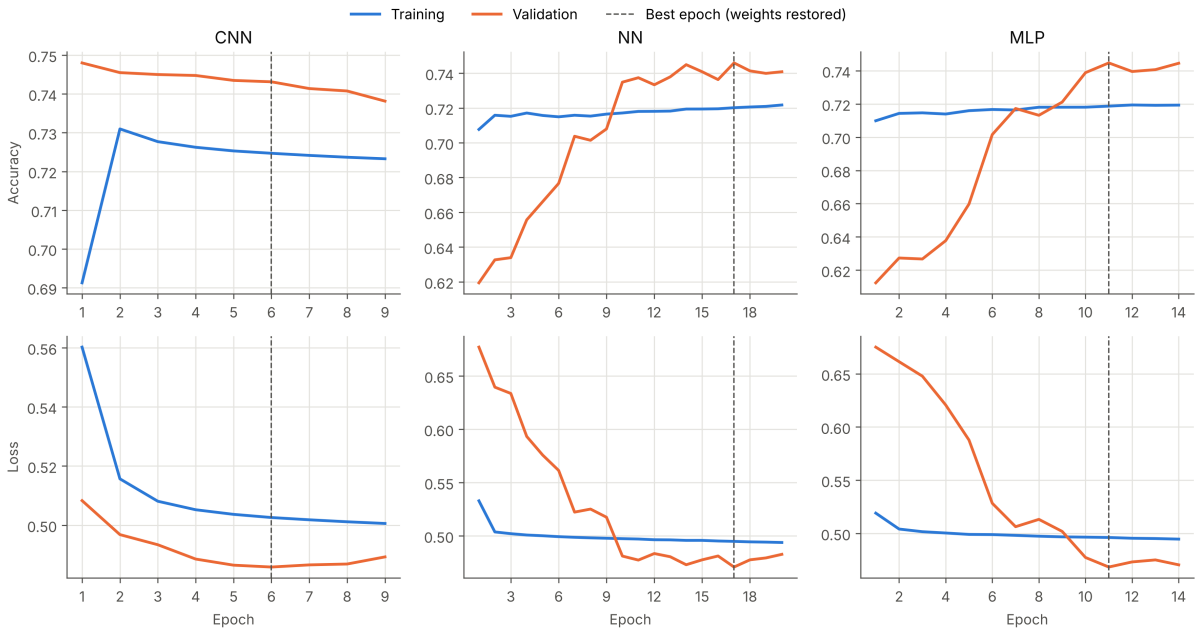


Figure 7: Training and validation accuracy (top) and loss (bottom) of the improved DL models; training stops early once the validation loss no longer improves, and the weights of the best epoch (dashed line) are restored.

Table 6: Five-fold cross-validation of the improved models on real records at the natural prevalence of CVD (9.3%); mean ± standard deviation over the folds, in %.

Model	Accuracy	Precision	Recall	F1 Score	AUC
DT	71.82 ± 1.92	21.84 ± 0.98	78.90 ± 2.17	34.18 ± 1.01	82.36 ± 0.36
SVM	72.16 ± 0.57	22.28 ± 0.36	80.65 ± 0.49	34.92 ± 0.45	83.53 ± 0.38
XGBoost	71.08 ± 1.00	21.88 ± 0.49	82.50 ± 1.22	34.58 ± 0.55	83.85 ± 0.35
LR	71.38 ± 1.68	21.94 ± 0.82	81.52 ± 1.72	34.56 ± 0.90	83.55 ± 0.38
CNN	71.41 ± 1.41	21.88 ± 0.73	81.04 ± 1.54	34.44 ± 0.84	83.25 ± 0.64
NN	70.79 ± 1.09	21.61 ± 0.52	81.95 ± 2.05	34.20 ± 0.61	83.37 ± 0.51
MLP	72.19 ± 2.54	22.29 ± 1.28	80.01 ± 2.49	34.83 ± 1.37	83.36 ± 0.38

among the nine features selected by correlation, which explains why using all features improves discrimination. Conversely, BMI, days of poor physical health and health care coverage are not significantly associated with CVD once the other variables are taken into account.

Some associations run against intuition: heavy alcohol consumption (OR 0.72) and higher income (0.97 per level) are associated with lower odds, and physical activity and daily vegetable consumption with slightly higher odds. Because BRFSS is cross-sectional and records past diagnoses, these are associations, not causal effects; they plausibly reflect reverse causation, as people often change their behaviour after a diagnosis of heart disease. The model is suited to identifying who is likely to have CVD, not to estimating the effect of modifying a risk factor.

5.6 Calibration

Class weighting improves recall, but it distorts the predicted probabilities. On the out-of-fold predictions, where 9.3% of the records have CVD, the improved models predict an average risk of 32–36% (Table 8), and their Brier scores (0.150–0.177) are worse than that of a constant prediction equal to the prevalence (0.084). The raw outputs therefore must not be read as individual risks. After the prior correction of Section 4.4, which uses no additional data, the mean predicted risk is 7.6–9.5% and the Brier score falls to 0.070–0.072, better than the constant

Table 7: Adjusted odds ratios (OR) of CVD from a multivariable logistic regression with all 21 features, fitted to the 179,024 real training records.

Variable	OR	95% CI	p
Stroke (yes)	2.61	2.47–2.76	<0.001
Male sex	2.13	2.05–2.21	<0.001
High cholesterol (yes)	1.84	1.77–1.91	<0.001
Cholesterol check in past 5 years (yes)	1.81	1.55–2.12	<0.001
High blood pressure (yes)	1.67	1.60–1.74	<0.001
Poorer general health (per level)	1.60	1.56–1.63	<0.001
Smoker (yes)	1.41	1.36–1.46	<0.001
Difficulty walking (yes)	1.34	1.29–1.41	<0.001
Age (per 5-year category)	1.28	1.27–1.29	<0.001
No doctor visit due to cost (yes)	1.24	1.17–1.32	<0.001
Diabetes (per level)	1.15	1.12–1.17	<0.001
Daily vegetables (yes)	1.06	1.02–1.11	0.005
Physical activity (yes)	1.06	1.02–1.11	0.003
Daily fruit (yes)	1.03	0.99–1.07	0.113
Education (per level)	1.02	1.00–1.04	0.078
Poor mental health (per day)	1.00	1.00–1.00	0.057
BMI (per unit)	1.00	1.00–1.00	0.695
Poor physical health (per day)	1.00	1.00–1.00	0.689
Income (per level)	0.97	0.96–0.97	<0.001
Health care coverage (yes)	0.95	0.87–1.05	0.320
Heavy alcohol consumption (yes)	0.72	0.65–0.78	<0.001

Table 8: Calibration of the improved models on the out-of-fold cross-validation predictions (223,781 records, observed prevalence 9.3%), before (raw) and after the prior correction for class weighting. A constant prediction equal to the prevalence has a Brier score of 0.084.

	Mean predicted risk (%)		Brier score	
Model	Raw	Corrected	Raw	Corrected
DT	35.9	9.2	0.177	0.072
XGBoost	35.4	9.0	0.170	0.070
LR	36.1	9.5	0.171	0.071
CNN	33.8	8.0	0.156	0.072
NN	32.5	7.6	0.150	0.072
MLP	32.6	7.7	0.151	0.072

prediction. The corrected calibration curves (Fig. 8) lie close to the diagonal for XGBoost; the MLP slightly underestimates and LR slightly overestimates the risk in the highest-risk group. With this correction, which leaves the AUC and the classifications unchanged, the models provide both a classification and a usable risk estimate.

5.7 Comparison with Existing Studies

Table 9 compares the proposed method with existing studies. A meaningful comparison requires the same dataset and an evaluation protocol that keeps the test data unseen, so the table is organised by dataset.

Same dataset (BRFSS 2015). Deng et al. [42] trained a LightGBM model on the same survey data and obtained an AUC of 0.811 with 90.6% accuracy. This accuracy was measured at the natural prevalence of about 10%, where a trivial classifier that always predicts "no CVD" already reaches 89.7% (reference row in Table 9), so it mainly reflects the class imbalance; our

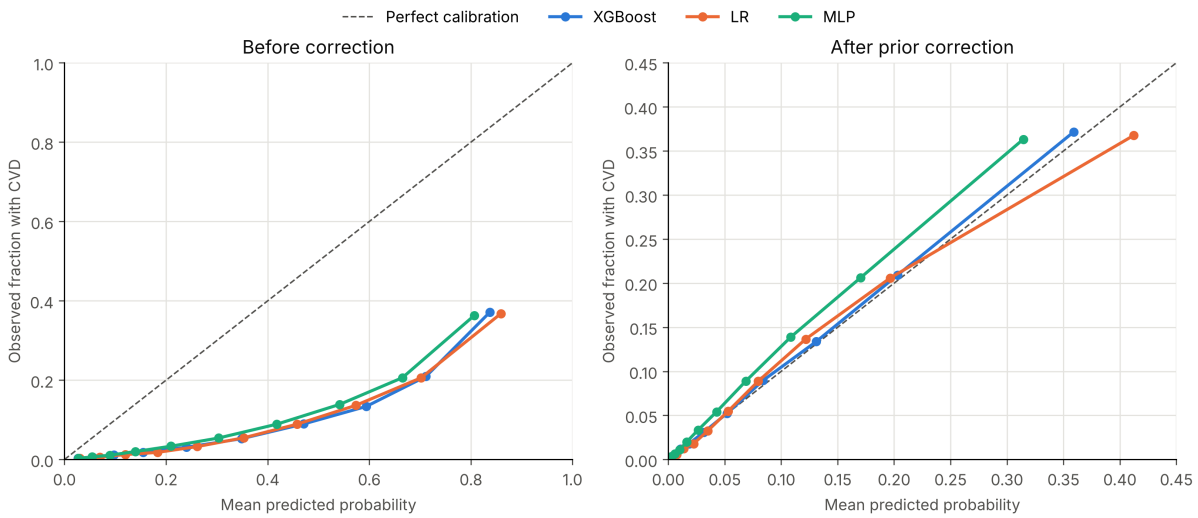


Figure 8: Calibration curves of the improved XGBoost, LR and MLP on the out-of-fold cross-validation predictions, before and after the prior correction for class weighting (ten quantile bins; note the different axis ranges).

accuracy, in contrast, is measured on a balanced test set, where the same trivial classifier scores 50%. On the AUC, which is comparable across both settings, all three proposed models in the table exceed Deng et al.: XGBoost reaches 0.840 with a recall of 82.8%, and logistic regression and the MLP reach 0.837 and 0.838. Deng et al.'s AUC of 0.811 lies below the lower 95% confidence bound of all three models (0.827–0.830; Table 5).

Cleveland dataset. Four studies in Table 9 use the clinical Cleveland data. Applied unchanged to this dataset, as described in Section 3.7, the improved pipeline reaches 80.1–81.3% accuracy and an AUC of 0.893–0.900 for SVM, XGBoost, LR and the three DL models (the depth-limited DT, whose leaf size of 50 records was chosen for the large BRFSS data, reaches 72.7%). These accuracies are slightly below the lowest published Cleveland result in the table (82.5%) and well below the highest (95.9%). The variation between folds, however, is large: the accuracy of LR on individual test folds of 59–60 patients ranges from 71.2% to 90.0%, so single-split results on this dataset are uncertain by several percentage points, and differences of a few points between studies are within this range.

Other datasets. The remaining studies use other clinical datasets (a CVD dataset, a Beijing hypertension cohort and the KNHANES survey) and are listed for completeness; their results are not directly comparable with ours.

Two factors explain why studies on clinical data report higher accuracy than any model trained on BRFSS. First, clinical measurements carry far more information about coronary disease than self-reported survey answers: on Cleveland, the same pipeline that reaches an AUC of 0.84 on BRFSS achieves an AUC of about 0.90. Second, accuracies estimated on a few dozen test patients vary by several percentage points between random splits, as our repeated cross-validation shows, and several studies do not state whether resampling or feature selection was performed before the data were split. Resampling before splitting is known to inflate performance estimates [41, 40]: when we resampled our data before splitting them, the same HS models reached 94.7–99.8% accuracy instead of 57.4–70.8%.

Table 9: Comparison with existing studies, grouped by dataset. All values in %. Accuracy, precision and F1 depend on the prevalence of CVD in the test data, so the AUC is the metric that is comparable across studies. Proposed results are leakage-free: BRFSS on the real, balanced held-out test set; Cleveland as mean ± standard deviation of 5×5-fold cross-validation.

Study	Dataset (n)	Method	Acc.	Prec.	Rec.	F1	AUC
Same dataset: BRFSS 2015 (self-reported survey)							
L. Deng et al. [42]	BRFSS 2015	LightGBM	90.61[a]	73.89[a]	–	–	81.11
Reference: always "no CVD"	BRFSS (229,781)	Majority class	89.68[a]	–	0.00	–	50.00
Proposed (improved)	BRFSS (229,781)	XGBoost	76.00	72.90	82.77	77.52	83.99
Proposed (improved)	BRFSS (229,781)	Logistic regression	76.00	73.93	80.33	77.00	83.70
Proposed (improved)	BRFSS (229,781)	MLP	75.47	72.37	82.40	77.06	83.79
Cleveland heart disease data (clinical measurements)							
S. K. Mohan et al. [28]	Cleveland (303)	Hybrid (HRFLM)	88.40	90.10	92.80	90.00	–
N. L. Fitriyani et al. [19]	Cleveland (303)	SMOTEENN + XGBoost	95.90	97.14	94.67	95.35	–
A. A. Almazroi et al. [37]	Cleveland (303)	DNN	82.49	–	87.24	84.41	–
J. P. Li et al. [18]	Cleveland (303)	SVM (Linear)	85.00	–	75.00	–	–
Proposed (improved)	Cleveland (297)	XGBoost	80.07 ± 5.1	82.48 ± 11.1	75.35 ± 10.5	77.61 ± 5.6	89.42 ± 4.2
Proposed (improved)	Cleveland (297)	Logistic regression	81.28 ± 4.7	84.63 ± 9.1	74.59 ± 10.0	78.47 ± 5.5	89.71 ± 3.8
Proposed (improved)	Cleveland (297)	MLP	80.53 ± 6.5	79.55 ± 10.1	80.58 ± 9.3	79.32 ± 6.2	89.38 ± 4.6
Other datasets (not directly comparable)							
M. Bhatt et al. [35]	CVD dataset	XGBoost	87.02	89.62	82.11	86.30	95.00
W. Chang et al. [36]	Beijing dataset	Hybrid XGBSVM	91.70	–	–	–	93.90
T. Amarbayasgalan et al. [38]	KNHANES (25,340)	Two DNNs	89.20	91.10	92.00	91.50	97.54

[a] Measured at the natural prevalence (≈10% CVD), where always predicting "no CVD" gives 89.7% accuracy; not comparable with accuracy on a balanced test set.

5.8 Limitations

The BRFSS data are self-reported and cross-sectional: the target records a past diagnosis of coronary heart disease or myocardial infarction, so the models identify individuals who resemble patients with a history of CVD rather than predict future events. Self-reported survey answers also carry less information than clinical measurements, which limits the attainable discrimination. The models were developed and tested on a single survey year without external validation; calibration was therefore assessed only internally, and the prior correction assumes that the prevalence in the target population matches that of the training data. The balanced test set makes accuracy comparable across models, but precision and F1 depend on prevalence, and the cross-validation results indicate the values to expect in a general population. The Cleveland evaluation used the hyper-parameters chosen for BRFSS without re-tuning, which favours a fair comparison over the best attainable score on that dataset.

6 CONCLUSION AND FUTURE WORK

This study combined hybrid sampling (SMOTEENN followed by ROS) with three DL models (CNN, NN, MLP) and three classical ML models (DT, SVM, XGBoost) for CVD prediction from the 2015 BRFSS survey. On 6,000 real records held out before any sampling, these models reach only 57.4–70.8% accuracy and miss most CVD cases, because hybrid sampling removes borderline records and duplicates the remaining minority records, leaving only the most clear-cut CVD cases in the training data.

We therefore proposed a leakage-free pipeline that uses all available features, handles class imbalance with class weights instead of synthetic records, regularises the models and tunes the decision threshold on a validation set. On real, unseen records it improves every model, to about 76% accuracy, 80% recall and an AUC of 0.84, values that five-fold cross-validation confirms. The technical observations of this study are:

- On real data, hybrid sampling did not improve discrimination: it lowered the AUC of XGBoost from 0.823 to 0.743. Class weighting and a tuned decision threshold address imbalance without altering the training data.

- Correlation-based selection of nine features discarded useful information; using all 21 features raised the AUC.
- Class weighting distorts predicted probabilities: the raw outputs overestimate the average risk three- to fourfold. A simple prior correction restores calibration (Brier score 0.070–0.072) without changing the AUC or the classifications, and should be applied whenever the models are used to estimate individual risk.
- A history of stroke, male sex, high cholesterol, high blood pressure and poorer general health are the strongest risk factors; as the data are cross-sectional, these are associations rather than causal effects.
- Model choice matters far less than data handling. Logistic regression, SVM, XGBoost and the three DL models all reach an AUC of 0.83–0.84, so simple and interpretable models are a sound choice for this task.
- Compared on equal terms, the improved pipeline exceeds the AUC of a published model on the same BRFSS data (0.840 vs. 0.811), and applied unchanged to the clinical Cleveland data it reaches an AUC of about 0.90. Higher accuracies reported on clinical datasets largely reflect more informative clinical measurements and small, variable test sets, and cannot be taken as evidence of a better method without a comparable, leakage-free evaluation.
- At the real prevalence of about 10%, a model that detects 80% of CVD cases still produces three to four false alarms for every true case (precision about 22%). Survey-based models are therefore suited to low-cost screening and risk stratification that triggers clinical follow-up, not to diagnosis.

Future work should validate the pipeline on external cohorts and later BRFSS survey years, assess calibration in external populations, and add clinical measurements (for example blood pressure readings, lipid profiles and ECG data), which are likely needed to move beyond the performance ceiling of self-reported data. Explainability methods such as SHAP values could make the models' risk factors transparent to clinicians, and threshold selection could be aligned with the costs of missed cases and false alarms in a specific screening programme. We also encourage authors of CVD prediction studies to report the point at which resampling is applied and to release their code, so that results can be verified and compared.

CODE AND DATA AVAILABILITY

The dataset used in this study is publicly available on Kaggle [30] and in the GitHub repository at `https://github.com/daniyalabdulah/Early-Detection-of-Heart-Disease-using-Deep-Learning`. The complete source code, which reproduces every table and figure in this paper, is available at `https://github.com/mohsinkhan85/Biomedical-Neurocomputing-Lab`.

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References

[1] World Health Organization. Cardiovascular Diseases. Webpage, 2023. Available at: `https://www.who.int/health-topics/cardiovascular-diseases` (Accessed on June 23, 2023).

[2] Colin D. Mathers and Dejan Loncar. Projections of global mortality and burden of disease from 2002 to 2030. *PLoS Med*, 3(11):e442, 2006. doi: 10.1371/journal.pmed.0030442.

[3] Mayo Clinic. Heart disease. Webpage, 2019. Available at: `https://www.mayoclinic.org/diseases-conditions/heart-disease/symptoms-causes/syc-20353118` (Accessed on June 23, 2023).

[4] Peter Libby. Mechanisms of acute coronary syndromes and their implications for therapy. *New England Journal of Medicine*, 368(21):2004–2013, 2013. doi: 10.1056/NEJMra1216063.

[5] Salim Yusuf, Steven Hawken, Stephanie Ôunpuu, Tony Dans, Alvaro Avezum, Fernando Lanas, Matthew McQueen, Andrzej Budaj, Prem Pais, John Varigos, and Liu Lisheng. Effect of potentially modifiable risk factors associated with myocardial infarction in 52 countries (the INTERHEART study): Case-control study. *The Lancet*, 364(9438):937–952, 2004. doi: 10.1016/S0140-6736(04)17018-9.

[6] Salim S. Virani, Alvaro Alonso, Eunsik J. Benjamin, Marcio S. Bittencourt, Clifton W. Callaway, April P. Carson, Alanna M. Chamberlain, Alexander R. Chang, Susan Cheng, Francesca N. Delling, et al. Heart disease and stroke statistics—2020 update: A report from the American Heart Association. *Circulation*, 141(9):e139–e596, 2020. doi: 10.1161/CIR.0000000000000757.

[7] Subhi J. Al'Aref, Kimberly Anchouche, Gaurav Singh, Patricia J. Slomka, Krzysztof K. Kolli, Amit Kumar, Mohit Pandey, Gabriel Maliakal, Alexander R. van Rosendael, Ashley N. Beecy, et al. Clinical applications of machine learning in cardiovascular disease and its relevance to cardiac imaging. *European Heart Journal*, 40(24):1975–1986, 2019. doi: 10.1093/eurheartj/ehy404.

[8] Aditi Gavhane, Gouthami Kokkula, Isha Pandya, and Kailas Devadkar. Prediction of heart disease using machine learning. In *2018 Second International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pages 1275–1278, 2018. doi: 10.1109/ICECA.2018.8474922.

[9] A. SheikAbdullah. A data mining model to predict and analyze the events related to coronary heart disease using decision trees with particle swarm optimization for feature selection. *International Journal of Computer Applications*, 55:49–55, 2012. doi: 10.5120/8779-2736.

[10] Geoffrey Currie, Kristina Hawk, Eric Rohren, Alanna Vial, and Ran Klein. Machine learning and deep learning in medical imaging: Intelligent imaging. *Journal of Medical Imaging and Radiation Sciences*, 50, 2019. doi: 10.1016/j.jmir.2019.09.005.

[11] M. Chlif, N. M. Ammar, N. Ben Said, L. Sergey, S. Ahmaidi, F. Alassery, and H. Hamam. Mechanism of dyspnea during exercise in children with corrected congenital heart disease. *Int. J. Environ. Res. Public Health*, 19(99), 2022. doi: 10.3390/ijerph19010099.

[12] N. M. Hussain, A. U. Rehman, M. T. B. Othman, J. Zafar, H. Zafar, and H. Hamam. Accessing artificial intelligence for fetus health status using hybrid deep learning algorithm (AlexNet-SVM) on cardiotocographic data. *Sensors*, 2022. doi: 10.3390/s22145103.

[13] A. A. Khan, A. A. Laghari, M. Shafiq, O. Cheikhrouhou, W. Alhakami, H. Hamam, and Z. A. Shaikh. Healthcare ledger management: A blockchain and machine learning-enabled novel and secure architecture for medical industry. *Human-centric Computing and Information Sciences*, 12, 2022. doi: 10.22967/HCIS.2022.12.055.

[14] Abid Ishaq, Saima Sadiq, Muhammad Umer, Saleem Ullah, Seyedali Mirjalili, Vaibhav Rupapara, and Michele Nappi. Improving the prediction of heart failure patients' survival using SMOTE and effective data mining techniques. *IEEE Access*, 9:39707–39716, 2021. doi: 10.1109/ACCESS.2021.3064084.

[15] Arpit Singh and Anuradha Purohit. A survey on methods for solving data imbalance problem for classification. *International Journal of Computer Applications*, 127(15):37–41, 2015. doi: 10.5120/ijca2015906677.

[16] Gustavo Batista, Ronaldo Prati, and Maria-Carolina Monard. A study of the behavior of several methods for balancing machine learning training data. *SIGKDD Explorations*, 6:20–29, 2004. doi: 10.1145/1007730.1007735.

[17] Liaqat Ali, Abdur Rahman, Aurangzeb Khan, MengChu Zhou, Adeeb Javeed, and Jude A. Onwuegbuzie. Efficient prediction of cardiovascular disease using machine learning algorithms with Relief and LASSO feature selection techniques. *IEEE Access*, 9:24505–24522, 2021. doi: 10.1109/ACCESS.2021.3053759.

[18] Jian Ping Li, Amin Ul Haq, Salah Ud Din, Jalaluddin Khan, Asif Khan, and Abdus Saboor. Heart disease identification method using machine learning classification in e-healthcare. *IEEE Access*, 8:107562–107582, 2020. doi: 10.1109/ACCESS.2020.3001149.

[19] Norma Latif Fitriyani, Muhammad Syafrudin, Ganjar Alfian, and Jongtae Rhee. HDPM: an effective heart disease prediction model for a clinical decision support system. *IEEE Access*, 8:133034–133050, 2020. doi: 10.1109/ACCESS.2020.3010511.

[20] Ghulab Nabi Ahmad, Hira Fatima, Shafi Ullah, Abdelaziz Salah Saidi, et al. Efficient medical diagnosis of human heart diseases using machine learning techniques with and without GridSearchCV. *IEEE Access*, 10:80151–80173, 2022. doi: 10.1109/ACCESS.2022.3165792.

[21] T. R. Ramesh, Umesh Kumar Lilhore, M. Poongodi, Sarita Simaiya, Amandeep Kaur, and Mounir Hamdi. Predictive analysis of heart diseases with machine learning approaches. *Malaysian Journal of Computer Science*, pages 132–148, 2022. doi: 10.22452/mjcs.sp2022no1.10.

[22] Muhammad Bilal Qureshi, Jason Glower, Daniel L. Ewert, and Steven C. Koenig. A novel idea to improve cardiac output of mechanical circulatory support devices by optimizing kinetic energy transfer available in forward moving aortic blood flow. *Cardiovascular Engineering and Technology*, 8(2):131–144, 2017. doi: 10.1007/s13239-017-0305-2.

[23] Javed Azmi, Muhammad Arif, Md Tabrez Nafis, M. Afshar Alam, Safdar Tanweer, and Guojun Wang. A systematic review on machine learning approaches for cardiovascular disease prediction using medical big data. *Medical Engineering & Physics*, page 103825, 2022. doi: 10.1016/j.medengphy.2022.103825.

[24] Vijendra Singh, Vijayan K. Asari, and Rajkumar Rajasekaran. A deep neural network for early detection and prediction of chronic kidney disease. *Diagnostics*, 12(1), 2022. doi: 10.3390/diagnostics12010116.

[25] Vijendra Singh and Divya Jain. A hybrid parallel classification model for the diagnosis of chronic kidney disease. *International Journal of Interactive Multimedia and Artificial Intelligence*, 2021. doi: 10.9781/ijimai.2021.10.008.

[26] Divya Jain and Vijendra Singh. A two-phase hybrid approach using feature selection and adaptive SVM for chronic disease classification. *International Journal of Computers and Applications*, 43:1–13, 2019. doi: 10.1080/1206212X.2019.1577534.

[27] Divya Jain and Vijendra Singh. A novel hybrid approach for chronic disease classification. *International Journal of Healthcare Information Systems and Informatics (IJHISI)*, 15(1):1–19, 2020. doi: 10.4018/IJHISI.2020010101.

[28] Senthilkumar Mohan, Chandrasegar Thirumalai, and Gautam Srivastava. Effective heart disease prediction using hybrid machine learning techniques. *IEEE Access*, 7:81542–81554, 2019. doi: 10.1109/ACCESS.2019.2923707.

[29] M. Kavitha, G. Gnaneswar, R. Dinesh, Y. Rohith Sai, and R. Sai Suraj. Heart disease prediction using hybrid machine learning model. In *2021 6th International Conference on Inventive Computation Technologies (ICICT)*, pages 1329–1333, 2021. doi: 10.1109/ICICT50816.2021.9358597.

[30] Alex Teboul. Heart Disease Health Indicators Dataset. 2015. `https://www.kaggle.com/datasets/alexteboul/heart-disease-health-indicators-dataset`.

[31] Centers for Disease Control and Prevention (CDC). Behavioral Risk Factor Surveillance System (BRFSS), 2022. Available at: `https://www.kaggle.com/datasets/cdc/behavioral-risk-factor-surveillance-system?select=2015.csv`.

[32] Centers for Disease Control and Prevention (CDC). Calculated variables in the 2015 data file of the BRFSS, 2015. Available at: `https://www.cdc.gov/brfss/annual_data/2015/pdf/2015_Calculated_Variables_Version4_08_10_17-508c.pdf` (Version 4, 08/10/17).

[33] Zahid Ullah and Mona Jamjoom. An intelligent approach for Arabic handwritten letter recognition using convolutional neural network. *PeerJ Computer Science*, 8:e995, 2022. doi: 10.7717/peerj-cs.995.

[34] Zahid Ullah and Mona Jamjoom. A deep learning for Alzheimer's stages detection using brain images. *Computers, Materials and Continua*, 74, 2022. doi: 10.32604/cmc.2023.032752.

[35] Chintan M. Bhatt, Parth Patel, Tarang Ghetia, and Pier Luigi Mazzeo. Effective heart disease prediction using machine learning techniques. *Algorithms*, 16(2), 2023. doi: 10.3390/a16020088.

[36] Wenbing Chang, Yinglai Liu, Xueyi Wu, Yiyong Xiao, Shenghan Zhou, and Wen Cao. A new hybrid XGBSVM model: Application for hypertensive heart disease. *IEEE Access*, 7:175248–175258, 2019. doi: 10.1109/ACCESS.2019.2957367.

[37] Abdulwahab Ali Almazroi, Eman A. Aldhahri, Saba Bashir, and Sufyan Ashfaq. A clinical decision support system for heart disease prediction using deep learning. *IEEE Access*, 11:61646–61659, 2023. doi: 10.1109/ACCESS.2023.3285247.

[38] Tsatsral Amarbayasgalan, Van-Huy Pham, Nipon Theera-Umpon, Yongjun Piao, and Keun Ho Ryu. An efficient prediction method for coronary heart disease risk based on two deep neural networks trained on well-ordered training datasets. *IEEE Access*, 9:135210–135223, 2021. doi: 10.1109/ACCESS.2021.3116974.

[39] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012. doi: 10.1145/2382577.2382579.

[40] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[41] Miriam Seoane Santos, Jastin Pompeu Soares, Pedro Henriques Abreu, Hélder Araújo, and João Santos. Cross-validation for imbalanced datasets: avoiding overoptimistic and overfitting approaches. *IEEE Computational Intelligence Magazine*, 13(4):59–76, 2018.

[42] Lang Deng, Kongjie Lu, and Huanhuan Hu. An interpretable LightGBM model for predicting coronary heart disease: Enhancing clinical decision-making with machine learning. *PLOS ONE*, 20(9):e0330377, 2025. doi: 10.1371/journal.pone.0330377.

[43] Andras Janosi, William Steinbrunn, Matthias Pfisterer, and Robert Detrano. Heart Disease [Dataset]. UCI Machine Learning Repository, 1989. doi: 10.24432/C52P4X.

[44] Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10:36, 2017. doi: 10.1186/s13040-017-0154-4.